12.7.6 Wrap-Up: Error Analysis

1. The number of campers who attend a local summer camp has increased from $1{,}938$ to $2{,}128$. The camp director calculates the percent of increase for an upcoming presentation. Her work is shown.

   $2{,}128 - 1{,}938 = 190$

   $190 \div 2{,}128 = 0.089$

   $0.089 = 8.9\%$

   Explain the camp director's mistake.

Unit 12, Lesson 8: Calculating Percent Error and Simple Interest

Benchmark

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.

Learning Targets

- ☐ I can find percent error.
- ☐ I can solve problems that involve simple interest.

12.8.1 Warm-Up: Would You Rather?

It's "Multiple Monday" and the more pizzas purchased, the greater the discount applied. If one pizza is ordered, the discount is 10%. If two pizzas are ordered, the discount is 20%, and for three pizzas, the discount goes up to 30%.

1. Would you rather order 3 pizzas on separate checks, or order 3 pizzas on the same check and split the cost?

2. Justify your answer using mathematics.

12.8.2 Exploration: Estimate vs. Actual

1. Instructions to care for a plant say, "Water with $\frac{3}{4}$ cup of water daily." The plant has been getting 25% too much water. How much water has the plant been getting? Include units.

2. The pressure on a bicycle tire is 63 psi. This is 5% higher than what the manual states is the correct pressure. What is the correct pressure? Include units.

3. The crowd at a sporting event is estimated to be $2{,}500$ attendees. The exact attendance is $2{,}486$ people. By what percentage is the estimated attendance off?

4. Discuss with your partner what strategies you used to solve questions $1-3$.

12.8.3 Guided Instruction: Percent Error

Percent error is the difference between the estimated number and the actual number as a percentage of the actual value.

Percent error can be used to describe any situation where there is a correct value and an incorrect value, and to describe the relative difference between them.

1. A teacher estimates there are approximately 600 students at his school, but in actuality there are 625 students enrolled.

   a. What is the difference between the teacher's estimate and the actual number of students enrolled at the school?

   b. Using the definition of percent error, finish filling out the ratio of the situation.

$$\left|\frac{\text{Expected Value} - \text{Actual Value}}{\text{Actual Value}}\right| = \left|\frac{\quad}{\quad}\right| = 0.\_\_\_\_$$

   c. The percent error in the teacher's estimate is _____%.

   d. Discuss with your partner why the percent error is positive. Summarize your discussion.

2. The front of a cereal box lists the net weight of the product is 21 ounces (oz). After further measurements, it is determined there are actually 23 oz of cereal in the box.

   a. What is the difference, in ounces, between the weight listed on the box and the measured weight?

   b. Find the percent error. Show your work.

12.8.4 Guided Practice: Percent Error

1. A store owner estimated she would have 125 customers in her store on its opening day. She underestimated the number of customers by 8%. How many customers visited the store on opening day?

2. To be labeled a jumbo egg, the egg is supposed to weigh 2.5 ounces (oz). Alana buys a carton of jumbo eggs and measures one of the eggs as 2.4 oz. What is the percent error?

3. A student estimates there to be 25.7 milliliters (mL) of a solution in his beaker in science class. He underestimated the amount by 12.5%. What was the actual amount of solution in the beaker? Round to the nearest thousandth.

4. Lia wanted to take advantage of the pizza special on "Multiple Monday." She estimates it will cost her $\$28$ for two large cheese pizzas after the discount is applied. The actual cost is $\$28.80$. What is the percent error? Round to the nearest whole percent.

12.8.5 Guided Instruction: Simple Interest

1. Kristina received $\$100$ in a birthday card and wants to put the money into her savings account, where she will earn an additional 2% each year it remains in her account. This is known as ***simple interest***.

> ***Simple interest*** is a method of computing interest. Interest is computed from the (original) principal alone, no matter how much money has accrued so far.

   a. How much is 2% of $\$100$?

   b. How much money will be in Kristina's savings account after one year if she makes no additional deposits or withdrawals after putting in her birthday money?

   c. Compare your answer with your partner and discuss how you determined the answer for Part B.

2. Suppose an investment of $\$100$ earns $\$3.50$ interest in one year.

   a. What investment amount will earn $\$61.60$ in two years at the same rate?

   b. Ask two other partner pairs to describe how they found the initial investment. Record their initial investment and write your summary of their process. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Initial Investment	My Summary of Their Reason	Initials

   c. Review the initial investments and reasons provided by other partner pairs. If necessary, fix any mistakes you and your partner made.

Simple Interest Formula

$$I = p \times r \times t$$

where I is the interest earned, p is the principal starting amount of money, r is the interest rate per year, and t is the length of the investment.

3. Tonio took out a six-year car loan for $\$15{,}600$, with a 3% simple interest rate over the life of the loan.

   a. How much interest, in dollars, will Tonio pay annually?

   b. How much will Tonio pay in interest over the course of **six** years?

   c. What is the total amount Tonio will pay back at the end of his six-year loan?

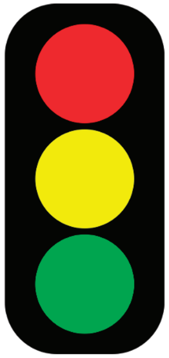
   d. Explain why the total amount Tonio will have paid back by the end of his loan is more than the loan amount he borrowed.

12.8.6 Wrap-Up: Reflection

In this unit, you have learned multiple concepts about percentages and proportional relationships, including those listed below.

- Solving real-world problems using proportions
- Solving discount, markup, tax, tip, and fees problems
- Finding percentage of increase and decrease
- Finding percent error
- Solving problems involving simple interest

1. Use the stoplight reflection below to share how you are feeling about each of the concepts at this point.



I still feel stuck on how to …

I'm starting to get the hang of …

I feel confident and ready to go with …

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